Does Outcome Sparsity Limit Uplift-Ranking Evidence?

Research memo summarizing a comparative evaluation of causal uplift estimators on CRITEO-UPLIFTv2.1. All reported statistics are computed from held-out test-set evaluation artifacts produced by the project's experimental evaluation pipeline; full analysis code and computational provenance are available in the accompanying reproducibility repository.

Executive Summary

The business problem. An advertiser wants to know which users an ad actually *changes the mind of* -- not just which users are likely to buy anyway. Ad budget spent on someone who would have converted regardless is wasted; the useful targeting signal is the *incremental* effect of showing the ad, not the raw likelihood of the outcome on its own.

Two ways to target. A simple approach ranks users by how likely they are to convert, without regard to whether they were shown an ad -- a *response model* (here, Response LightGBM; formally, this predicts `P(Y|X)`, the outcome probability given the user's features). A more sophisticated approach instead tries to estimate each user's actual incremental response to the ad -- an *uplift*, or *causal*, model (here, T-Learner, X-Learner, and Causal Forest; formally, these estimate the conditional average treatment effect `tau(X)`). This memo asks whether the added complexity of the causal approach measurably outranks the simpler response model, using CRITEO-UPLIFTv2.1.

What we found. We compare all four estimators on their ability to rank users by incremental treatment effect on **conversion**, the primary business outcome. Conversion is rare (0.29% prevalence), and under it, Response LightGBM's point-estimate ranking edges out all three causal estimators, but a paired bootstrap shows this gap is **not statistically distinguishable from resampling noise** on any of the three ranking metrics checked: Qini and AUUC (two different ways of scoring how much better than random targeting a model's ranking is, summed across the whole ranked list) and uplift@10% (the incremental effect captured if only the top 10% of ranked users were targeted). This result is genuinely inconclusive, not evidence that the causal estimators fail to add value.

As a robustness check, we repeat the identical comparison on **visit**, a much denser behavioral outcome (4.66% prevalence, ~16x conversion's rate) on the *same* test rows. There, Causal Forest's point-estimate lead over Response LightGBM **is** statistically distinguishable from noise on all three metrics. Read together, the two results suggest that the ambiguity under conversion is consistent with an outcome-sparsity limit on this comparison's statistical power, rather than evidence that no ranking difference exists between the response baseline and the causal estimators.

What this means in practice. On conversion, the outcome that matters for the business, this analysis does not give statistical grounds to prefer either the response baseline or the causal estimators -- the honest position today is that the two approaches are not distinguishable in expected ranking performance on this outcome, not that one has been shown to beat the other. This is not a claim that Causal Forest is the better model in general, that visit substitutes for conversion as the business objective, or that the bootstrap check establishes a causal mechanism -- see Limitations, and Section 8 for what a decision maker can and cannot conclude from this analysis.

1. Research Question

An advertiser wants to know for which users an ad *changes* behavior, not merely which users are likely to act regardless of exposure. Response LightGBM answers the latter question, `P(Y|X)`, ignoring treatment entirely. T-Learner, X-Learner, and Causal Forest instead estimate the conditional average treatment effect, `tau(X) = E[Y(1) - Y(0) | X]`, and rank users by the effect ad exposure is estimated to have had on them. This memo asks whether that additional causal machinery produces a measurably better ranking than the simpler response baseline on held-out data -- and whether any answer to that question is stable across two related but distinct outcome definitions on the same population.

2. Data and Outcome Definitions

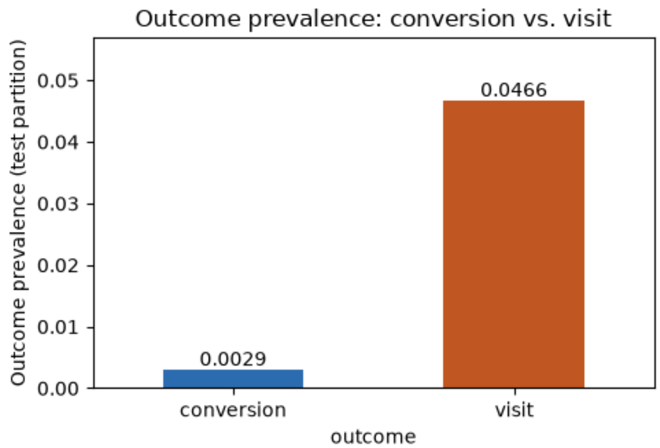
Both outcomes are evaluated on the identical seeded 70/15/15 train/validation/test partition of CRITEO-UPLIFTv2.1 (`n_test = 2,096,939`); only which column is read as `Y` differs.

Table 1 -- Outcome definitions and test-partition prevalence

Outcome	Definition	Prevalence (test)	Treatment rate
`conversion` (primary)	User completes a purchase following ad exposure	0.2917%	85%
`visit` (robustness check)	User visits the advertiser's site following ad exposure	4.6631%	85%

`visit` is ~16.0x more prevalent than `conversion` on the same rows. Every conversion is preceded by a visit, so visit sits on the same causal path, one step upstream. A rarer outcome gives every estimator fewer positive labels per treatment arm to learn from and to be evaluated against -- this is the basis for treating visit as a check on whether conversion's conclusions are limited by label sparsity, addressed directly in Sections 5-6.

Figure 1 -- Outcome prevalence: conversion vs. visit



Prevalence and treatment-rate figures are computed directly from the held-out test partition.

3. Experimental Design

Table 2 -- Estimator objectives

Estimator	Estimates	Role
Response LightGBM	`P(Y|X)`	Non-causal targeting baseline
T-Learner	`tau(X)`	Two independent per-arm outcome models, differenced
X-Learner	`tau(X)`	Cross-fitted correction of T-Learner's imbalanced-arm bias
Causal Forest	`tau(X)`	Honest random forest splitting directly on treatment-effect heterogeneity

("Cross-fitted" means each user's prediction comes from a model that never saw that user during training, avoiding the bias of a model grading its own homework. "Honest" means a tree decides *where* to split the data on one subset and estimates the effect size on a separate subset, so the same rows are never used to both pick and score a split.)

All four are scored on the test partition only (never touched during fitting or model selection) with **Qini above the theoretical random reference** (how much better than random targeting the model's ranking does, cumulated as more users are targeted) as the primary ranking statistic, **AUUC** (Area Under the Uplift Curve -- a companion ranking score built the same way but weighted differently across the ranking) as a secondary view of the same ranking, and **uplift@K** (the incremental effect captured if only the top K% of ranked users were targeted) at five coverage levels.

Section 5's significance check compares **Causal Forest against the Response LightGBM baseline** specifically, using the same pairing under both outcomes. T-Learner and X-Learner's point estimates are reported in Section 4 for completeness but are not carried into the significance check; the rationale for this specific pairing is given in Robustness and Limitations, below.

Full evaluation-protocol and modeling-methodology documentation is available in the project README.

4. Results

4.1 Conversion Outcome

Table 3 -- Model comparison, conversion outcome (test partition)

Model	Objective	Qini above random	AUUC above random	uplift@10%
Response LightGBM	`P(Y|X)`	806.24	942.03	0.00832
Causal Forest	`tau(X)`	769.58	897.73	0.00821

Model	Objective	Qini above random	AUUC above random	uplift@10%
X-Learner	`tau(X)`	544.11	636.21	0.00732
T-Learner	`tau(X)`	369.65	430.46	0.00634
Random (reference)	--	47.64	56.09	0.00117

All four models rank users better than random targeting. Response LightGBM has the highest point estimate on every metric shown; Causal Forest is the closest causal-model runner-up. Whether this point-estimate gap is real is addressed in Section 5, not here -- a leaderboard by construction always has a top row.

Figure 2 -- Qini above random by model and outcome (conversion panel)

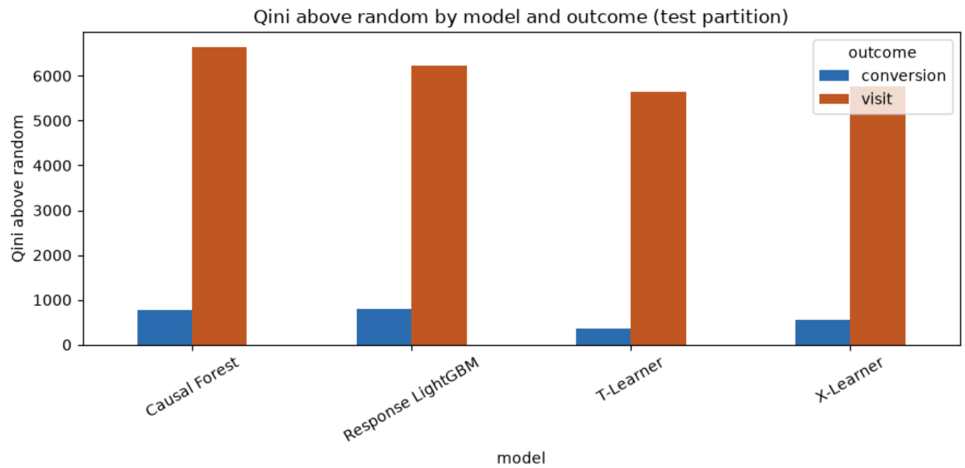
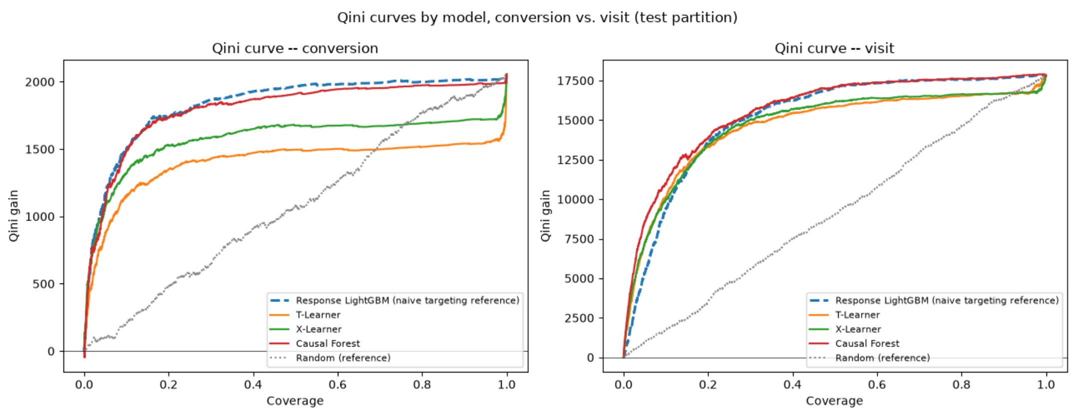


Figure 3 -- Qini curves, conversion panel



Figures 2-3 show both outcomes together.

Source: held-out test-set evaluation artifacts from the experimental evaluation pipeline (conversion outcome).

4.2 Visit Robustness Check

Table 4 -- Model comparison, visit outcome (test partition)

Model	Objective	Qini above random	AUUC above random	uplift@10%
Causal Forest	`tau(X)`	6642.20	7746.52	0.06052
Response LightGBM	`P(Y|X)`	6226.73	7280.12	0.05219
X-Learner	`tau(X)`	5759.00	6723.06	0.05499
T-Learner	`tau(X)`	5653.94	6591.47	0.05592
Random (reference)	--	186.67	219.93	0.00976

Under visit, Causal Forest's point estimate leads Response LightGBM's on every metric shown -- the ordering of the top two models inverts relative to Section 4.1. This table is presented as a check on the conversion result's reliability, not as a second, independent business finding: visit is not the outcome the business optimizes for.

(Figures 2-3 above already show the visit panel alongside conversion.)

Source: held-out test-set evaluation artifacts from the experimental evaluation pipeline (visit outcome).

5. Statistical Evidence

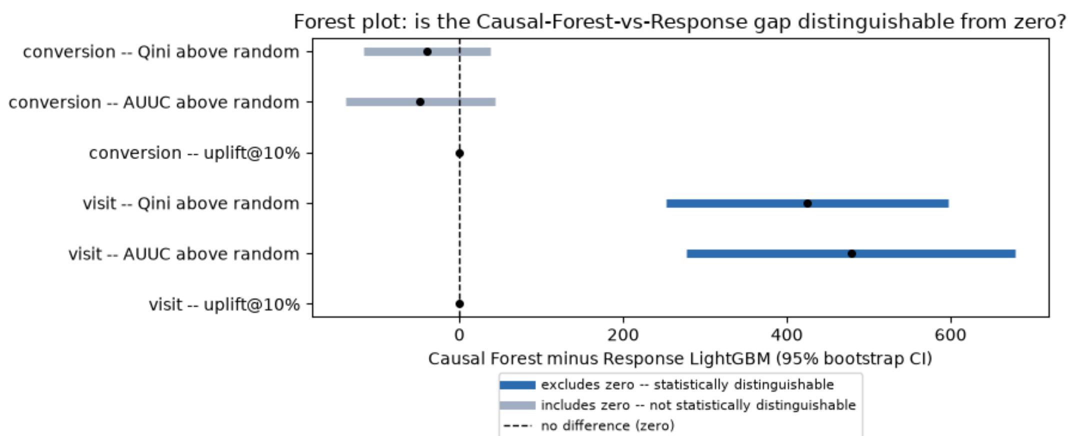
A point-estimate leaderboard always produces a "winner," even when two models are statistically indistinguishable. We use a paired bootstrap (500 resamples of the test rows with replacement; the same resample index applied to both models on each draw) to compute a 95% CI -- a confidence interval, the range that would contain the true gap in about 95 of 100 repeats of this resampling procedure -- on the **Causal Forest minus Response LightGBM** gap (evaluated model minus Response baseline), per outcome, for the three metrics shown above. Positive values indicate higher uplift-ranking performance for Causal Forest relative to the Response baseline; negative values indicate higher performance for the Response baseline.

Table 5 -- Paired bootstrap 95% CI, Causal Forest minus Response LightGBM

Outcome	Metric	95% CI (Causal Forest - Response)	Conclusion
conversion	Qini above random	[-117.68, 37.85]	includes zero
conversion	AUUC above random	[-139.48, 42.90]	includes zero
conversion	uplift@10%	[-0.00079, 0.00059]	includes zero
visit	Qini above random	[252.30, 596.53]	excludes zero
visit	AUUC above random	[277.28, 678.75]	excludes zero
visit	uplift@10%	[0.00370, 0.01249]	excludes zero

Under conversion, all three intervals include zero: the data cannot rule out that the true gap between Causal Forest and Response LightGBM is zero. Under visit, all three intervals sit entirely above zero, i.e., in Causal Forest's favor: the observed gap is unlikely to be explained by sampling variation alone. Because Qini, AUUC, and uplift@10% are constructed from the same ranking and the same cumulative outcome sums, these three results should be read as **consistent with one another**, not as three independent confirmations -- no correction for multiple comparisons was applied, and none was needed to reach that more modest reading.

Figure 4 -- Forest plot: is the Causal-Forest-vs-Response gap distinguishable from zero?



Source: paired bootstrap significance analysis from the experimental evaluation pipeline.

6. Discussion

The two results, read together, suggest that outcome sparsity -- not model choice -- is a plausible binding constraint on what this particular comparison's statistical power can resolve. The same estimators, fit and scored the same way, on the same rows, produce an inconclusive gap under a rare outcome and a resolvable one under a ~16x denser outcome on the identical partition. This is consistent with a rare-event variance explanation: conversion gives every estimator very few positive labels per arm to learn from and to be scored against, which widens the bootstrap CI regardless of whether a true ranking difference exists underneath it.

This explanation should not be treated as proven. It assumes visit is a denser view of a related estimation problem, rather than a genuinely different treatment-effect structure that happens to diverge from conversion's for unrelated reasons -- no artifact in this project directly measures whether visit-ranked and conversion-ranked uplift scores agree with each other, so that assumption is unverified (see Limitations). The conversion result should therefore be read as **inconclusive**, not as evidence that the causal estimators offer no benefit over the response baseline: a confidence interval that includes zero is consistent with both "no real difference" and "a real difference too small or too noisy to detect at this sample size," and the data here cannot distinguish between those.

T-Learner and X-Learner trail both Response LightGBM and Causal Forest on point estimates under both outcomes. This memo does not draw a conclusion about meta-learner estimators in general from that observation -- it is reported for completeness, and no bootstrap check was run on either of these two models against the others.

7. Limitations

- **Offline evaluation only.** Qini/AUUC are computed against the historical A/B test's logged outcomes; no online experiment confirms that acting on either model's ranking would reproduce the measured incremental effect.
- **The sparsity explanation in Section 6 is an inference, not a verified fact.** No artifact measures whether visit-ranked and conversion-ranked uplift scores are correlated; it remains possible that visit and conversion reflect partially different treatment-effect structures rather than one problem viewed at two densities.
- **Bootstrap coverage is partial.** Only the Causal-Forest-vs-Response-LightGBM gap was tested, on three (correlated) metrics, per outcome -- not T-Learner vs. X-Learner, nor any other pairwise comparison.
- **Resample-count and seed sensitivity were not checked.** The bootstrap procedure is deterministic given a fixed seed, but how much the reported CIs -- particularly conversion's, whose bounds sit only moderately from zero -- would shift under a different seed or resample count was not tested.
- **Causal Forest's categorical representation is coarser** than the LightGBM-based estimators' (frequency-capped top-8 one-hot vs. full-cardinality native categorical splits), a memory/runtime tradeoff the comparison does not adjust for.
- **Not a validated causal mechanism.** `f0`-`f11` are anonymized with no known business meaning, and the predicted CATE (the conditional average treatment effect `tau(X)` defined in Section 1, under its standard abbreviation) is not a true individual treatment effect -- both potential outcomes are never observed for the same user, so no PEHE (Precision in Estimation of Heterogeneous Effect, the standard accuracy check against true individual causal effects) against ground truth is reported. The bootstrap establishes statistical distinguishability of a ranking-metric gap; it does not establish or validate a causal mechanism.
- **Specific to this dataset, these implementations, one hyperparameter setting each.** A valid conclusion has the form "under outcome A, estimator X ranked better than estimator Y *here*," not a universal claim about either an outcome or a method.

8. Conclusion

Under conversion, the primary business outcome, Response LightGBM's point-estimate ranking lead over Causal Forest is not statistically distinguishable from resampling noise on any metric checked -- the comparison is inconclusive at this sample size, not a demonstrated win for either the response baseline or the causal estimators. The visit robustness check, run on the identical rows with a much denser outcome, provides evidence that this inconclusiveness is consistent with an outcome-sparsity limit on statistical power, since the same comparison becomes resolvable once given a denser signal. This is not a claim that Causal Forest is universally superior, that visit replaces conversion as the business objective, or that either result establishes a causal mechanism -- see Limitations for what this evidence does and does not support.

Practical implication. For a team choosing a targeting approach today, this analysis does not provide statistical grounds to prefer the causal estimators over the simpler response baseline on conversion, the metric that matters for the business -- nor does it provide grounds to prefer the response baseline over them. Neither approach should be presented as proven better on conversion from this evidence alone. If resolving that ambiguity matters, the indicated next step is gathering more conversion-labeled data or extending the observation window -- what the visit robustness check suggests would help is a denser outcome signal, not a different model choice.

Robustness and Limitations

A careful reader -- a technical reviewer, or an adjacent-field researcher checking this memo's methodology -- may reasonably raise the six questions below. Each is answered here directly, with a pointer to where the same point is discussed in the main text.

1. **Is the sparsity explanation just an assumption, not a proven fact?** Yes, and the memo says so directly: Section 6 states it as an inference and an unverified assumption, not a settled fact, and Section 7 restates it as a named limitation.
2. **Why Response LightGBM vs. Causal Forest specifically, and is that pairing cherry-picked?** No -- the pairing is fixed identically across both outcomes; it is not chosen after seeing which model leads: Causal Forest is the runner-up by point estimate under conversion

and the leader under visit, and the same two models are tested either way. Causal Forest is used as the causal-side comparator (rather than T-/X-Learner) because it is the most architecturally distinct of the three causal estimators -- it splits directly on treatment-effect heterogeneity rather than differencing two separately-fit outcome models, as described in Section 3.

3. **Are 500 bootstrap resamples enough, and is the result seed-sensitive?** This is an open limitation, stated in Section 7: no evidence exists to rebut the concern, so the memo does not claim more precision than was actually checked.
4. **Doesn't testing three correlated metrics inflate the chance of a false "significant" result?** Section 5 explicitly frames the three metrics as consistent with one another rather than independent confirmations, and states no multiple-comparisons correction was applied or needed under that framing.
5. **Is the secondary (visit) result being given more weight than it deserves?** No -- the Executive Summary and Section 8 both state the conversion result first and frame the visit result as informing the conversion result's reliability, not as a co-equal second finding.
6. **Is "robustness check" a precise term for what the visit comparison does?** This memo uses "robustness check" throughout and deliberately avoids the more loaded term "sensitivity analysis" for the visit comparison.

Each of the six questions above has either a stated resolution in the main text or an explicit, honestly-flagged limitation in Section 7.